

A disproof of Conjecture 00000002310

Base sizes of almost simple groups are unbounded

1 The conjecture

Conjecture 00000002310 reads (English version): “Definition: The base size $b(G)$ is the minimal size of a base with trivial pointwise stabilizer. Conjecture: The supremum of b over almost simple groups is 7 (attained by the action of $\mathrm{Sp}_8(2)$); the complete exception table for $b = 6$ or 7 contains exactly 3 groups.” The Chinese version has the same content.

Reading. The statement is a conjunction, and we refute its first conjunct: $\sup b = 7$ over almost simple groups. The class in the text is *all* almost simple permutation groups. The text does not restrict to “non-standard” actions or exclude the natural actions of alternating groups. We use the narrowest standard version of this class: finite almost simple groups acting faithfully and primitively on a finite set. A larger class, for example one without primitivity, only increases the supremum.

2 Definitions (as in Lean)

Definition 1. Let G act on Ω . A finite set $B \subseteq \Omega$ is a *base* if the only $g \in G$ fixing every point of B is $g = 1$. The *base size* $b(G)$ is the least cardinality of a base (`baseSize`, an infimum in $\mathbb{N}$).

Definition 2. G is *almost simple* (`IsAlmostSimple`) if it has a normal subgroup T that is a nonabelian simple group and has trivial centralizer $C_G(T) = 1$. Then conjugation embeds $T \cong \mathrm{Inn}(T) \leq G \leq \mathrm{Aut}(T)$.

Definition 3. `almostSimpleBaseSizes` is the set of values $b(G) \in \mathbb{N} \cup \{\infty\}$ over finite groups G that are almost simple and act faithfully (`FaithfulSMul`) and primitively (`Mathlib’s IsPreprimitive`) on a finite set Ω .

3 The alternating groups

Let $|\Omega| = n$ and let A_n act naturally on Ω .

Lemma 4. A set B with $|B| \leq n - 3$ is not a base.

Proof. The complement of B contains three distinct points a, b, c . The 3-cycle $\sigma = (ab)(bc)$ is even and fixes every point of B , but $\sigma(c) = a \neq c$. $\square$

Lemma 5. For any $a, b \in \Omega$, the complement $\Omega \setminus \{a, b\}$ is a base.

Proof. If $g \in A_n$ fixes every point outside $\{a, b\}$, then its support lies in $\{a, b\}$, so it has size 0, 1 or 2. Size 1 is impossible for a permutation. Size 2 makes g a transposition, which is odd. Hence the support is empty and $g = 1$. $\square$

Theorem 6. *For $n \geq 2$, $b(A_n) = n - 2$ in the natural action.*

Proof. Take $a \neq b$. Lemma 5 gives a base of size $n - 2$, and Lemma 4 shows that every base has size at least $n - 2$. $\square$

Lemma 7. *For $n \geq 5$, A_n is almost simple, and its natural action is faithful and primitive.*

Proof. Mathlib proves that A_n is simple for $n \geq 5$ (`alternatingGroup.isSimpleGroup`). Take $T = A_n$ itself. T is nonabelian: for distinct a, b, c, d , the elements $x = (a\ b)(a\ c)$ and $y = (a\ b)(a\ d)$ satisfy $xy(d) = a$ and $yx(d) = b$. The centralizer of T is the center, which is a normal subgroup. It is not all of T , because x and y do not commute, so by simplicity it is trivial. The action is faithful because A_n is a group of permutations. It is primitive for $n \geq 3$ (`alternatingGroup.isPreprimitive_of_three_le_card` in Mathlib). $\square$

Theorem 8. *The supremum of b over finite almost simple groups in faithful primitive actions is ∞ . In particular it is not 7. Concretely, A_{10} acting on 10 points is almost simple and primitive with $b = 8 > 7$.*

Proof. For each N , the group A_{N+5} lies in the class by Lemma 7 and has base size $N + 3 > N$ by Theorem 6. $\square$

Remark 9. Theorem 8 uses only the natural action of A_n . If the intended class is restricted to non-standard actions, which excludes these natural actions, our counterexample does not apply.

4 Lean 4 formalization

The file `lean/Conjecture2310/Basic.lean` (namespace `C2310`) imports only Mathlib. It contains:

- `IsBase, baseSize, IsAlmostSimple, almostSimpleBaseSizes;`
- `not_isBase` (Lemma 4), `isBase_compl_pair` (Lemma 5), `baseSize_alternating` (Theorem 6, for any finite type with at least 2 elements), `isAlmostSimple_alternating` (Lemma 7);
- `conjecture_2310_false: sSup almostSimpleBaseSizes = 7 and sSup almostSimpleBaseSizes  $\neq$  7;`
- `conjecture_2310_false_A10:  $A_{10}$  on  $\text{Fin } 10$  is almost simple and primitive, with baseSize = 8.`

The file has no `sorry` and no `native_decide`. `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.

5 Relation to the earlier submission

PR 222 (orionsheep) made the same mathematical point but was not merged. The reviewer wrote: “conjecture_refuted is $\neg(7 \geq 8)$, decide trivia with no groups in Lean, and the substantive part does not prove $b = 8$: `seven_points_fail` exhibits only ONE non-base 7-subset, whereas $b \geq 8$ requires every 7-subset to fail — that universal step is verified only in `reproduce.py`, and S_9 ’s almost-simplicity is prose.” This submission formalizes the missing pieces. It defines bases, base size and almost simplicity on actual group actions. It proves in Lean that *every* set of at most $n - 3$ points fails to be a base. It proves almost simplicity and primitivity of A_n in Lean. It gets the whole infinite family $b(A_n) = n - 2$, not one instance.